

# Potential changes in rainfall erosivity under climate change in southeastern United States

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## Abstract

Climate change will amplify erosion rates as erosive power of rainfall will likely increase due to change in rainfall characteristics (e.g., energy, intensity, duration as well as frequency). According to Intergovernmental Panel on Climate Change Fourth Assessment report (IPCC AR4), it is projected that by mid-21st Century, the rainfall across southeastern US will both increase and decrease in intensity, which will substantially affect rainfall erosivity. Few studies have estimated the impact of climate change (e.g., rainfall intensity) on rainfall erosivity across US and around the world. However, previously published erosion indices have discrepancies due to differences in methodologies (e.g., primarily omission of small and low rainfall intensity) adopted in those studies. Therefore, the objective of this study was to estimate change in erosion indices for the period 2030-2059 using the benchmark rainfall indices established for southeastern region of US. Hourly precipitation data were retrieved from NA-CORDEX under Representative concentration pathways (RCP) 8.5. Results on change in erosion indices as a result of climate change will be presented.

# Potential changes in rainfall erosivity under climate change in southeastern United States

**Potential changes in rainfall erosivity under climate change in southeastern United States**  
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**1. Background and Objectives**

- Study the average temperature is expected to be between 1.5 to 5.4°C by 2100.

**2. Study Site and Data**

**Observed rainfall:**  
 15-year rainfall for 129 stations (1975-2010) from NCEP/NCAR.

**Data screening:**  
 1. 20.21 (7088) approach estimates number of years and months per year to rain screening (Srivastava and Srivastava, 2019).  
 2. 187 total 576 stations passed the screening method.

**3. Research Methodologies**

**STEP-1: Bias-Correction**

- Simplified representation of physical laws.
- Quantile Delta mapping (QDM)

$$\hat{x} = x_{ex} \frac{F_x^{-1}(F_{ex}(x_{ex}))}{F_{ex}^{-1}(F_{ex}(x_{ex}))}$$

where,  
 $\hat{x}$  = Bias-Corrected rainfall  
 $x_{ex}$  = Observed  
 $F_x$  = Cumulative distribution probability function (CDF)  
 $F_{ex}$  = Observed  
 $F_x$  = Climate model  
 $F_{ex}$  = Observed  
 $F_x$  = Proposed future scenario

**STEP-2: Generation of 15-min precipitation**

- Modified stochastic method (Gnanadesikan et al., 2010; Wilson et al., 2011)

**4. Result and Discussions**

**1. Bias-correction:**

**5. Summary and Acknowledgments**

- Generated 15-min projected rainfall improved over hourly data.
- Annual rainfall accuracy is likely to increase spatial coverage.
- More reliable projected rainfall accuracy.

**LIMITATIONS:**

- Projected characteristics may not sufficiently represent observed characteristics.
- Estimation of rainfall erosivity is applicable only for Southeastern US.

**Keywords:**

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# 1. BACKGROUND AND OBJECTIVES

- Earth's average temperature is expected to be warmer between 1.1 to 5.4°C by 2100.

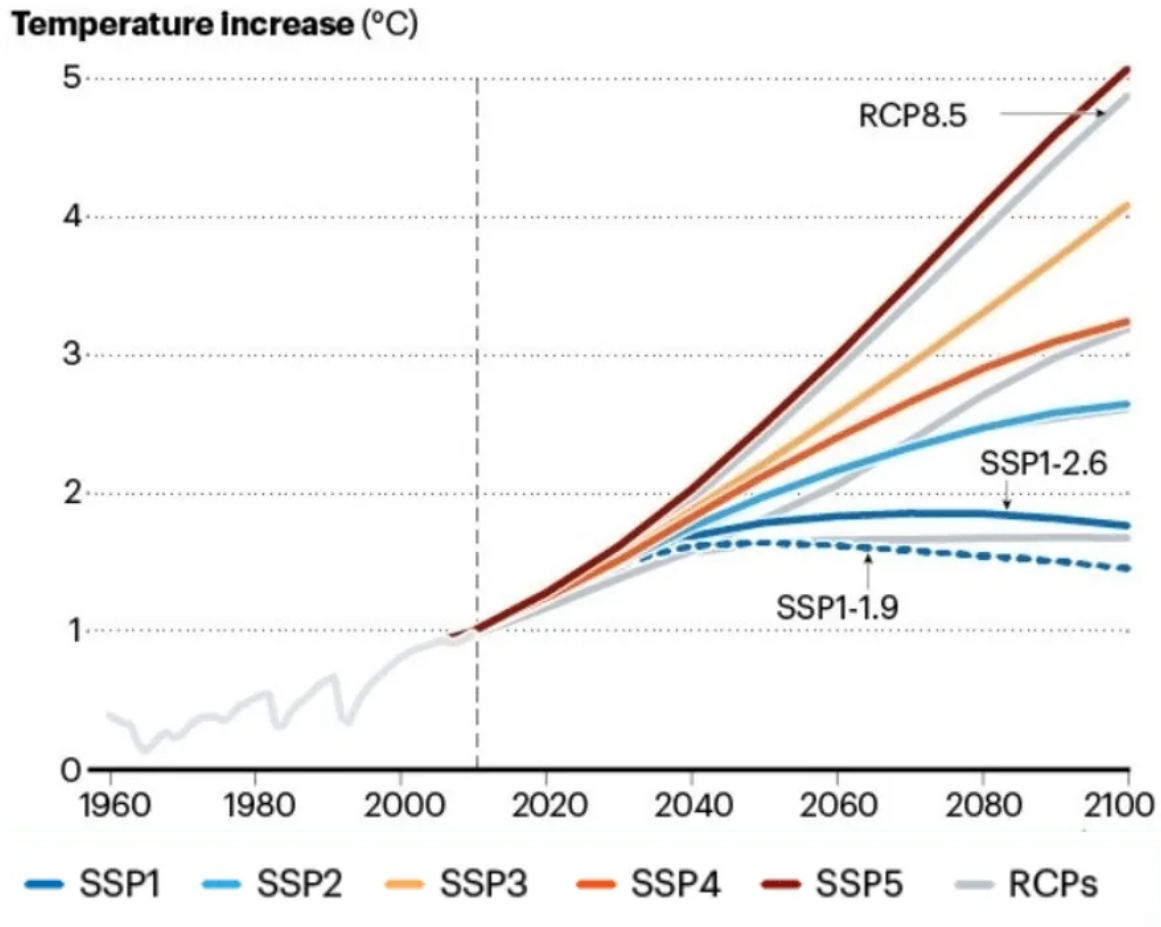


Fig 1: Projected temperature under different climate scenarios by 2100. Source: Jeff Tollefson (2020) (<https://www.nature.com/articles/d41586-020-01125-x>)

- Change in rainfall characteristics with climate change ( e.g., Intensity, duration, and frequency).  
For example: According to the **Clausius-Clapeyron relation**, precipitation intensity can be increased by 6-7% per 1°C.
- Southeastern, US has the greatest potential impact on rainfall erosivity than other parts of the country with 2000 to  $>10,000 \left( \frac{MJ \cdot mm}{ha \cdot h \cdot yr} \right)$ .
- Need to quantify a reliable projected potential of rainfall to erode soil (Rainfall Erosivity, R).

[VIDEO] [https://res.cloudinary.com/amuze-interactive/video/upload/vc\\_auto/v1638645506/agu-fm2021/2E-05-49-38-11-60-DA-D7-90-36-24-07-0E-0F-64-9F/Video/ParallelCandidAnteater\\_aufjm5.mp4](https://res.cloudinary.com/amuze-interactive/video/upload/vc_auto/v1638645506/agu-fm2021/2E-05-49-38-11-60-DA-D7-90-36-24-07-0E-0F-64-9F/Video/ParallelCandidAnteater_aufjm5.mp4)

Fig 2: Soil erosion by water. Source: TheGoodKid (<https://www.youtube.com/watch?v=N8C9OaBRW2g&fs=1&modestbranding=1&rel=0&showinfo=0>)

v=N8C9OaBRW2g&fs=1&modestbranding=1&rel=0&showinfo=0)

## Research Problems

1. Unavailability of high-resolution precipitation (daily rainfall led upto 30% under-prediction of R).

2. No reliable projected rainfall erosivity due to discrepancies in methodologies of previous studies (e.g. limitation of rainfall intensity to 76 mm/h, omission of rainfall < 2.54 mm).

## Solutions

1. Generation of 15-min precipitation using modified stochastic method.

2. Benchmarked reliable rainfall erosivity approach.

***Narration:***

## 2. STUDY SITE AND DATA

### **Observed rainfall:**

15-min rainfall for 578 stations (1971-2010) from NCEI-NOAA. (<http://www.ncei.noaa.gov/access/metadata/landing-page/bin/iso?id=gov.noaa.ncdc:C00505>)

### **Data screening:**

1. 20.11 (YY.MM) approach: minimum number of years and months per year to pass screening (McGehee and Srivastava, 2018).
2. 187 out of 578 stations passed the screening method.

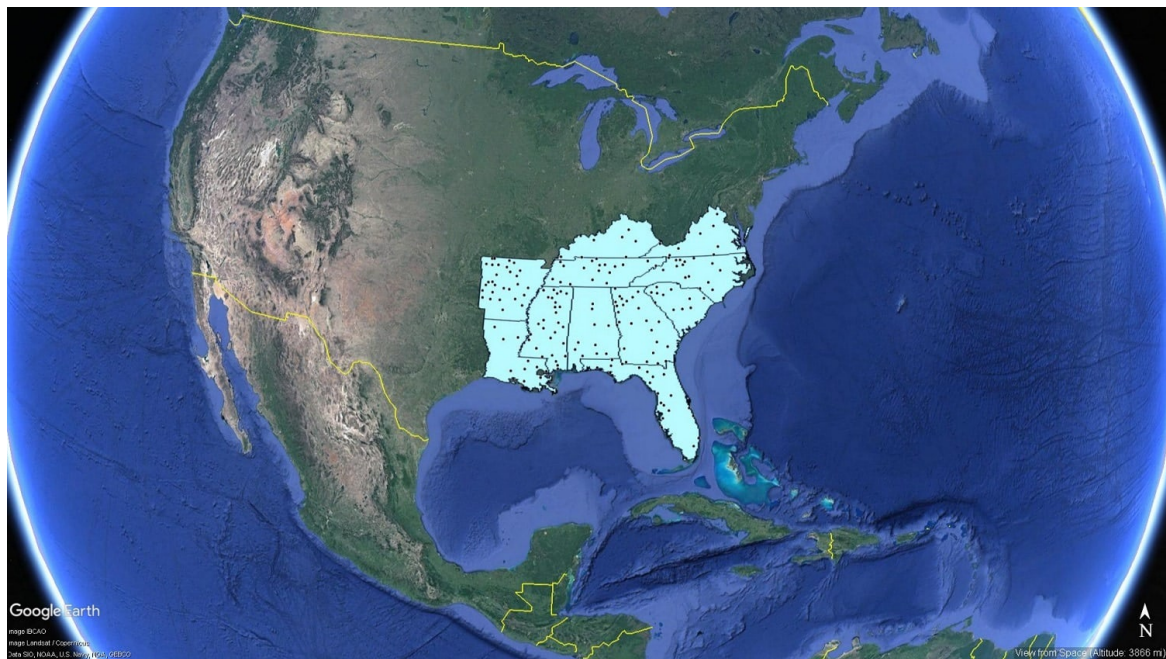


Fig 3: Locations of 187 observed precipitation stations after 20.11 screening.

### **Climate models:**

| Sl. No | Acronym            | Regional climate model                    | Contributing institution                                            |
|--------|--------------------|-------------------------------------------|---------------------------------------------------------------------|
| 1      | CANESM2_CANRCM4    | Canadian Regional Climate Model version 4 | Canadian Earth System Model                                         |
| 2      | HadGEM2-ES.WRF     | Weather Research and Forecasting          | Hadley Centre Global Environment Model version 2 Earth system model |
| 3      | GFDL-ESM2M.WRF     | Weather Research and Forecasting          | Earth System Model – Geophysical Fluid Dynamics Laboratory          |
| 4      | MPI-ESM-LR. RegCM4 | Regional Climate Model version 4          | Max Planck Institute for Meteorology Earth System Model LR          |
| 5      | MPI-ESM-LR.WRF     | Weather Research and Forecasting          | Max Planck Institute for Meteorology Earth System Model LR          |

Table 1. Description of climate models from NA-CORDEX (<https://www.earthsystemgrid.org/search/cordexsearch.html>).

**NOTE:** Hist =1970-1999 , rcp8.5= 2030-2059, Precipitation frequency= 1 hour & spatial resolution=50 Km<sup>2</sup>.

**Narration:**

### 3. RESEARCH METHODOLOGIES

#### **STEP-1: Bias-Correction.**

- Simplified representation of physical laws.
- **Quantile-Delta mapping (QDM):**

$$\hat{x} = x_m \frac{F_o^{-1}(F_{m,p}(x_{m,p}))}{F_{m,h}^{-1}(F_{m,p}(x_{m,p}))}$$

where,

$\hat{x}$  = Bias-Corrected rainfall

$x$  = Rainfall

$F$  = Cumulative distribution probability function (CDF),

$O$  = Observed,

$m$  = Climate model,

$h$  = Historical, and

$p$  = Projected/future scenarios.

#### **STEP-2: Generation of 15-min precipitation.**

- Modified stochastic method (Socolofsky et al. (2010) (<http://ascelibrary.org/doi/10.1061/%28ASCE%291084-0699%282001%296%3A4%28300%29>), Mirhosseini et al. (2013) (<http://link.springer.com/article/10.1007/s10113-012-0375-5>)).

[VIDEO] [https://res.cloudinary.com/amuze-interactive/image/upload/f\\_auto,q\\_auto/v1638489526/agu-fm2021/1b-73-70-31-f8-84-34-8b-0b-6a-ba-54-cc-5d-1b-97/image/gif\\_yb1uck.mp4](https://res.cloudinary.com/amuze-interactive/image/upload/f_auto,q_auto/v1638489526/agu-fm2021/1b-73-70-31-f8-84-34-8b-0b-6a-ba-54-cc-5d-1b-97/image/gif_yb1uck.mp4)

Fig 4: Stochastic generation of 15-min rainfall using cumulative distribution function (CDF).

where,

$D_t$  = Depth of measured/hourly rainfall

$a$  = Probability corresponding to  $D_t$

$U_i$  = Uniformly distributed random number between 0 and a

$D_i$  = rainfall event depth corresponding to  $U_i$

#### **STEP-3: Rainfall Erosivity(R).**

- Modified Agricultural Handbook (AH)–537 using recommendations from [Agricultural Handbook–703](https://agris.fao.org/agris-search/search.do?recordID=XF2015047686) (<https://agris.fao.org/agris-search/search.do?recordID=XF2015047686>) and [Mcgregor et.al \(1995\)](https://elibrary.asabe.org/abstract.asp?aid=27921) (<https://elibrary.asabe.org/abstract.asp?aid=27921>).

##### **Recommendations:**

1. High-resolution precipitation (fixed-intensity)
2. Includes all storms
3. Energy equations of AH-537
4. Maximum Intensity should not limited.
5. No regression between R and rainfall.

##### **a) Rainfall energy per unit depth:**

$$e \left( \frac{MJ.mm}{ha.h} \right) = 0.119 + 0.0873 * \log_{10}(I)$$

##### **b) Storm Erosivity:**

$$E_s \left( \frac{MJ.mm}{ha.h} \right) = \left[ \sum_{t=1}^p (e \cdot P) \right] I_{30}$$

##### **c) Average annual erosivity:**

$$R\left(\frac{MJ.mm}{ha.h.y}\right) = \frac{1}{n} \sum_{j=1}^n \left[ \sum_{k=1}^m (E_s)_k \right]$$

where,

I= rainfall intensity  $\left(\frac{mm}{h}\right)$ ,

P=rainfall depth ( $mm$ ),

I<sub>30</sub> = maximum 30-minute rainfall intensity.

p = number of time segments in the event,

t=single time interval,

j = index of the number of years used to produce an average,

n = number of years used to obtain average R,

k = index of the number of storms in each year, and

m = number of storms in each year.

**Narration:**

## 4. RESULT AND DISCUSSIONS

### 1. Bias-correction:

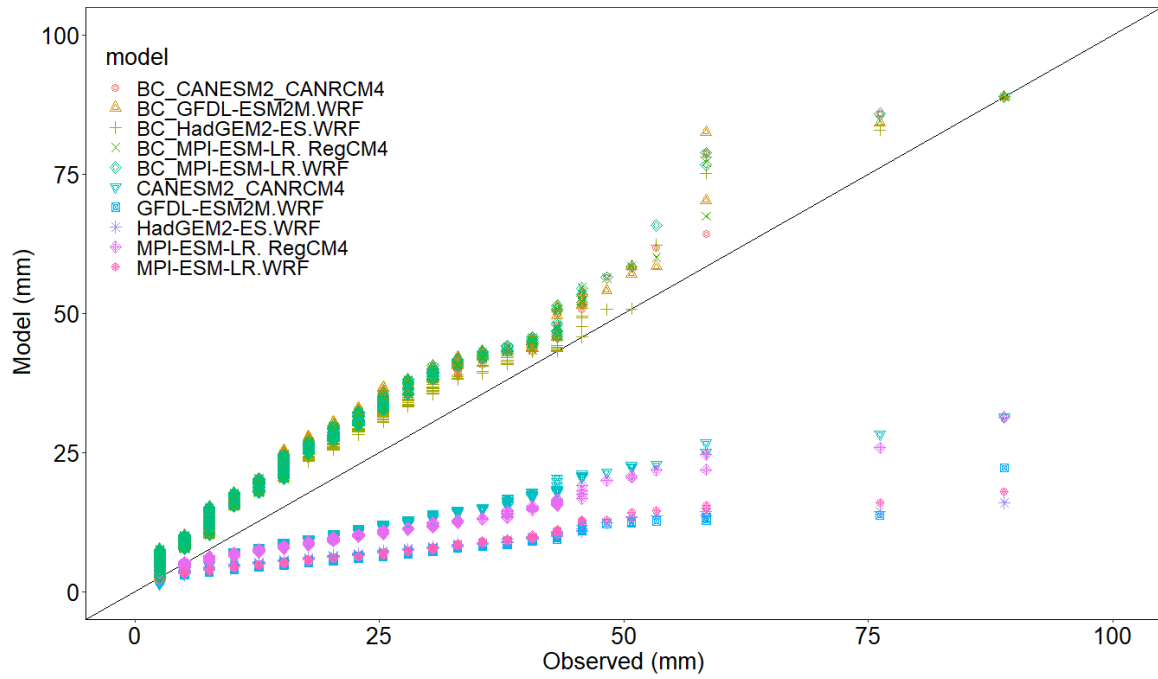


Fig 5: Quantile-quantile plot for biased-corrected and historical precipitation under different climate models (**NOTE:** BC= bias-corrected).

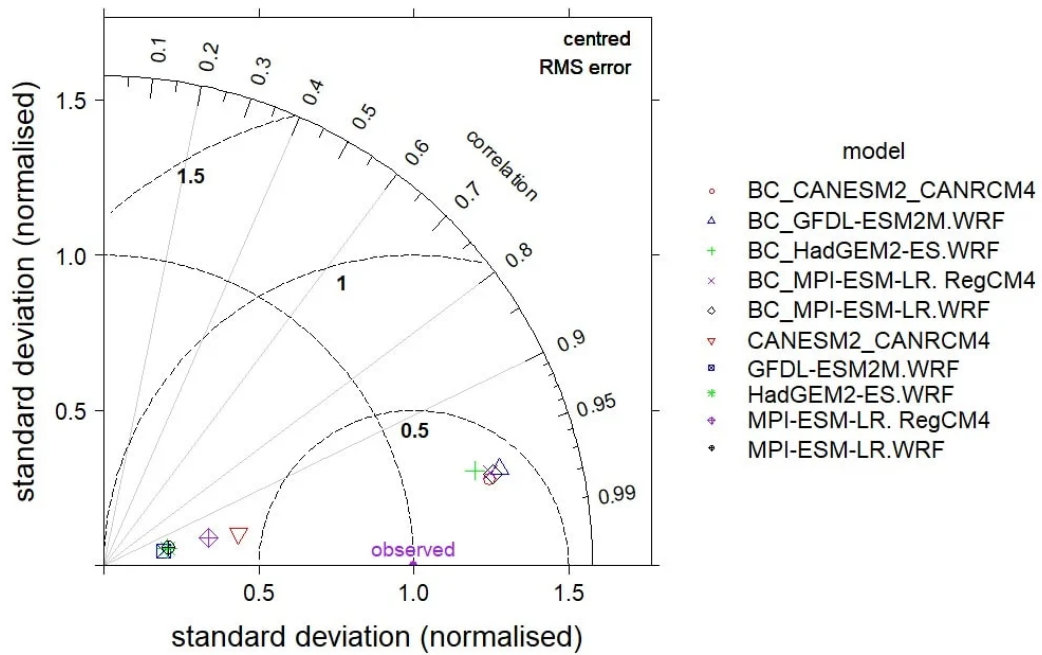


Fig 6: Taylor diagram for biased-corrected and historical hourly precipitation under different climate models.



## 2. Disaggregation:

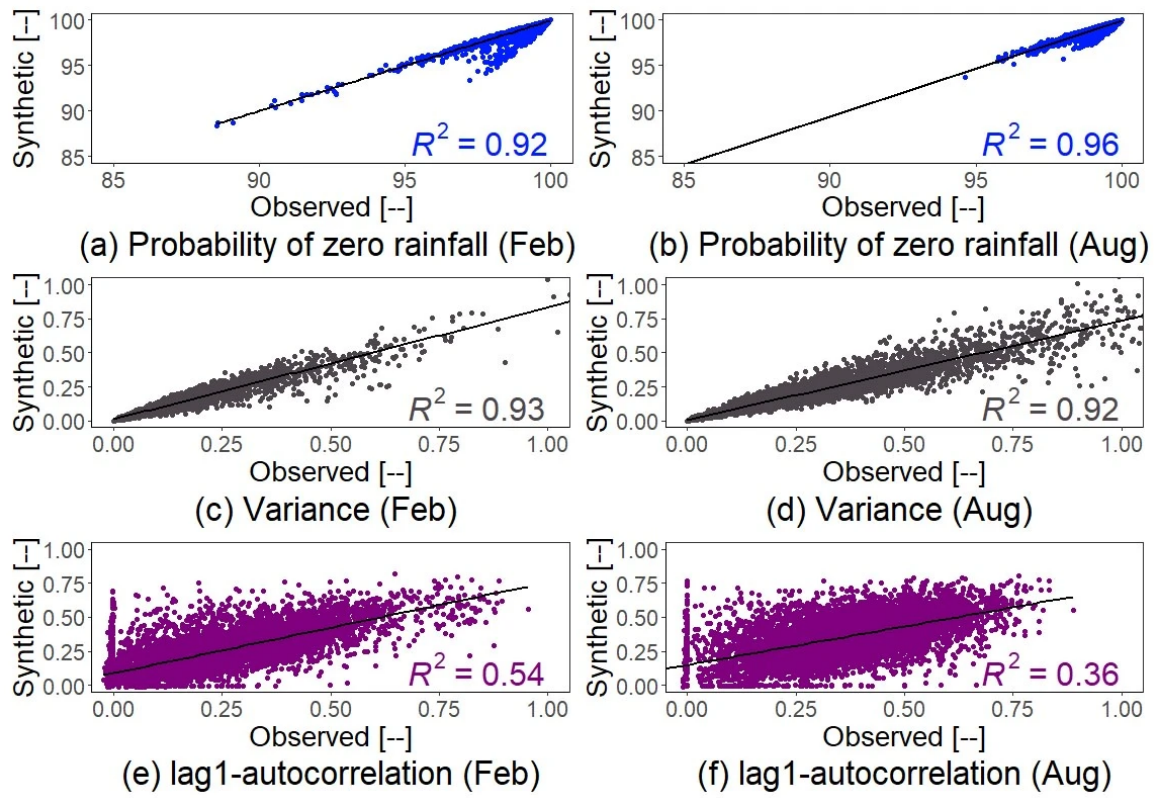


Fig 7: Statistical comparison between observed and disaggregated precipitation for winter (February) and summer (August).

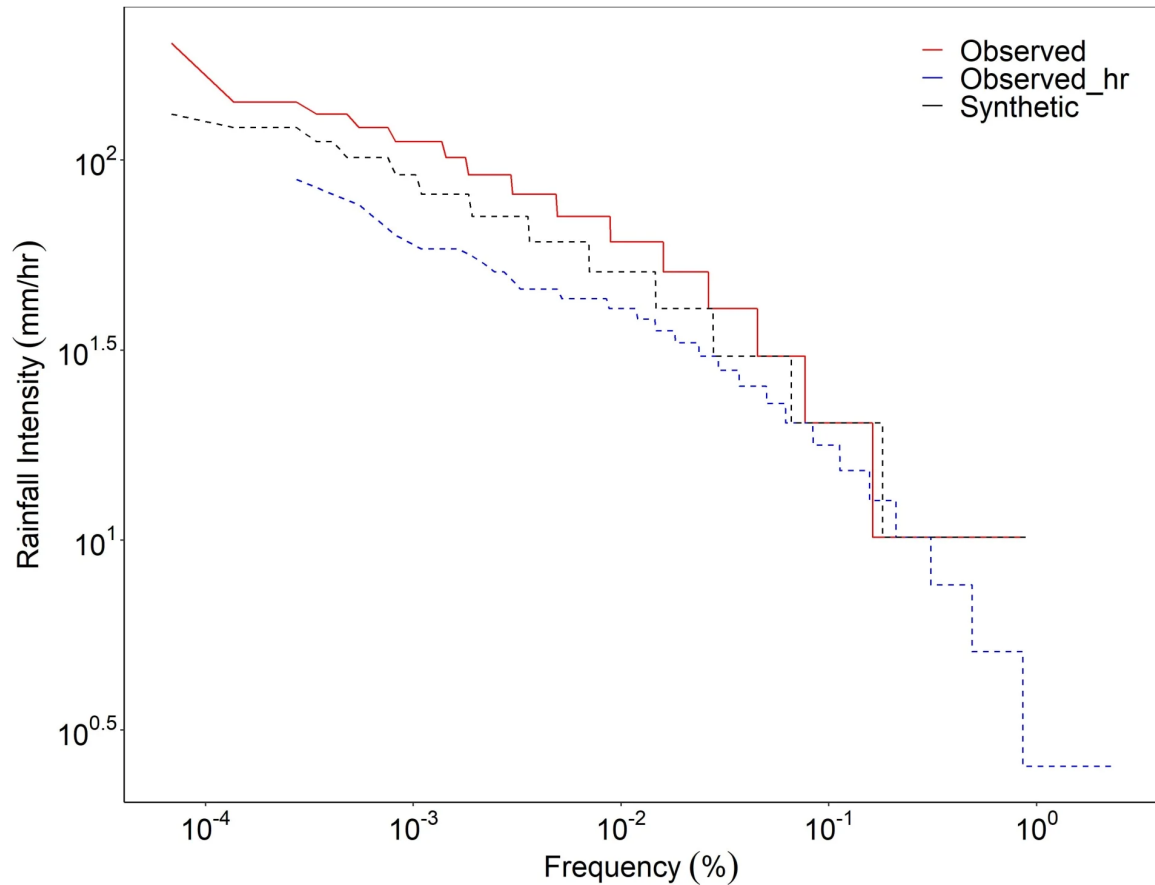


Fig 8: Comparison of precipitation intensity between observed(15-min), hourly-observed, and synthetic (15-min) precipitation.

## 3. Rainfall Erosivity:

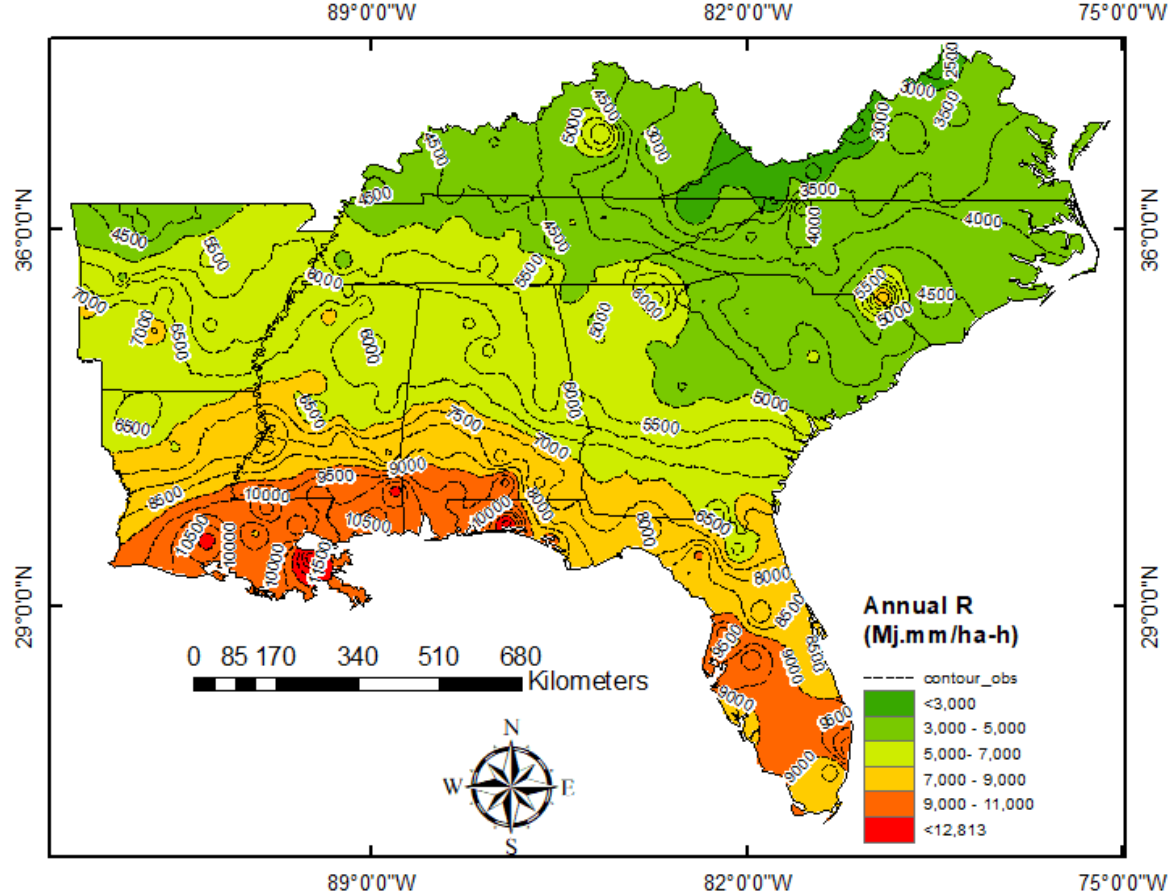


Fig 9: Observed annual rainfall erosivity (1971-2010) for Southeastern, USA.

[VIDEO] [https://res.cloudinary.com/amuze-interactive/image/upload/f\\_auto,q\\_auto/v1638739142/agu-fm2021/2e-05-49-38-11-60-da-d7-90-36-24-07-0e-0f-64-9f/image/ci\\_argf8d.mp4](https://res.cloudinary.com/amuze-interactive/image/upload/f_auto,q_auto/v1638739142/agu-fm2021/2e-05-49-38-11-60-da-d7-90-36-24-07-0e-0f-64-9f/image/ci_argf8d.mp4)

Fig 10: Projected annual average rainfall erosivity (2030-59) under different climate models.

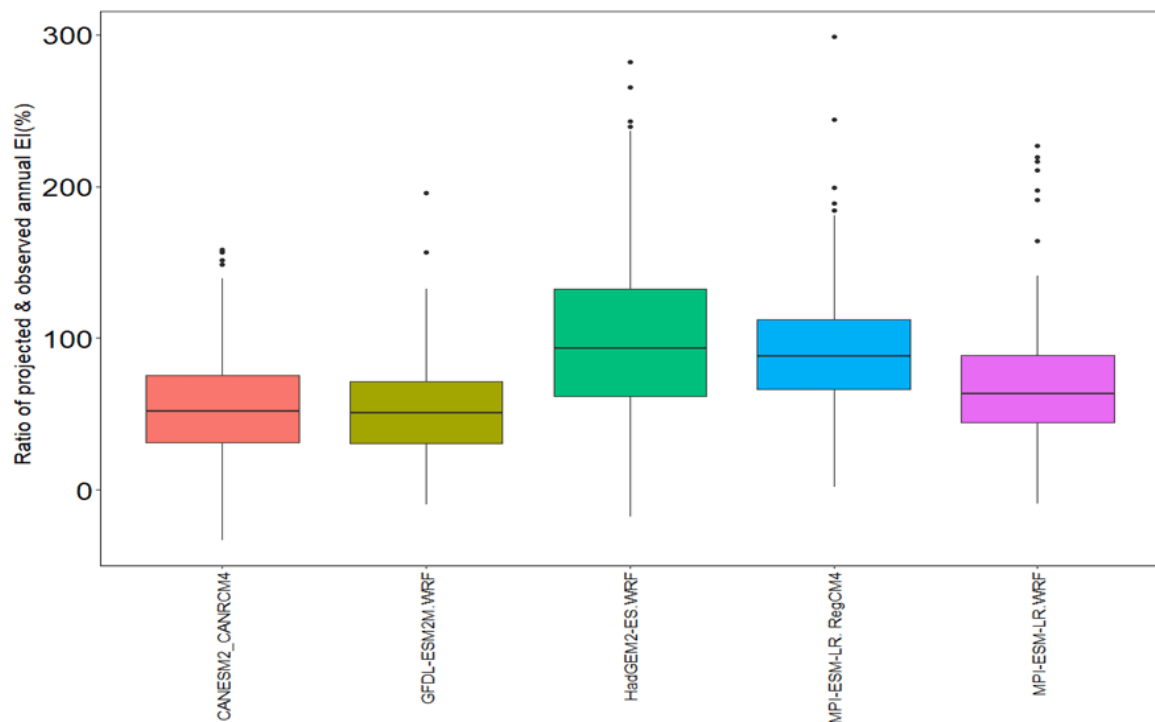


Fig 11: Boxplot for the projected relative change in annual rainfall erosivity (in %) with reference to 1970-2010 under different climate models.

Narration:

## 5. SUMMARY AND ACKNOWLEDGMENT

- Generated 15-min projected rainfall improved over hourly data.
- Annual rainfall erosivity is likely to increase spatial coverage.
- More reliable projected rainfall erosivity.

### **LIMITATIONS:**

- Rainfall characteristics also might not sufficiently represent observed characteristics.
- Estimation of rainfall erosivity is applicable only for Southeastern, US.

### **Narration:**

### **ACKNOWLEDGMENT:**

- Auburn University Hopper Cluster for assistance with this work.
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(<https://www.researchgate.net/profile/Takhellambam-Bijoychandra-Singh>).

# ABSTRACT

Climate change will amplify erosion rates as erosive power of rainfall will likely increase due to change in rainfall characteristics (e.g., energy, intensity, duration as well as frequency). According to Intergovernmental Panel on Climate Change Fourth Assessment report (IPCC AR4), it is projected that by mid-21<sup>st</sup> Century, the rainfall across southeastern US will both increase and decrease in intensity, which will substantially affect rainfall erosivity. Few studies have estimated the impact of climate change (e.g., rainfall intensity) on rainfall erosivity across US and around the world. However, previously published erosion indices have discrepancies due to differences in methodologies (e.g., primarily omission of small and low rainfall intensity) adopted in those studies. Therefore, the objective of this study was to estimate change in erosion indices for the period 2030-2059 using the benchmark rainfall indices established for southeastern region of US. Hourly precipitation data were retrieved from NA-CORDEX under Representative concentration pathways (RCP) 8.5. Results on change in erosion indices as a result of climate change will be presented.

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